

## 1.9.3–1.9.5 Wireframing, Figma and Prototypes

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CS XII

Chapter 1: Computer Systems (HCI)

Lecture 13 of 16

## 1.9.3–1.9.5 Wireframing, Figma and Prototypes

### LEARN IT

**1.9.3 Wireframing** is creating a basic visual outline or blueprint of a user interface **before** the final design. It plans the structure without focusing on colours or detailed graphics.

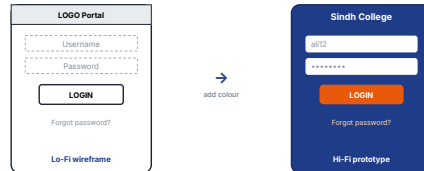
#### Low-fidelity wireframes (Lo-Fi)

A simple black-and-white sketch showing basic layout (boxes and lines). Used early to test and share ideas quickly. Focuses purely on structure and flow.

Lo-Fi sketch → Add details → Hi-Fi prototype

#### High-fidelity prototypes (Hi-Fi)

A detailed version that looks almost exactly like the final product. Includes real colours, fonts, images and interactive clickable buttons. Used for final user testing.



A school-portal login: Lo-Fi boxes and labels on the left; a coloured, realistic Hi-Fi prototype on the right.

### 1.10 What is Figma?

**Figma** is a popular, free, web-based tool used to design and collaborate on UI and prototypes. It works directly in a web browser — you do not have to install a heavy program on the college lab PCs.

#### Creating a low-fidelity design in Figma

1. **Sign up:** access the Figma workspace online.
2. **Create frames:** select the Frame tool (for example an iPhone-sized frame) to act as your screen canvas.
3. **Add shapes:** draw basic rectangles for buttons and circles for avatars.
4. **Add text:** use the text tool to label buttons (for example, "Sign Up").
5. **Duplicate and edit:** copy the frame to create the next screen (for example, the Home Screen).

### 1.9.5 Prototype design

A **prototype** is an early interactive model used to test design ideas **before writing actual code**. In Figma you use the Prototype tab to link screens together.

- Set a **Trigger** (for example, "On Tap").
- Set an **Action** (for example, "Navigate to Home Screen").

Real users can then click through the fake app to see if the navigation makes sense.

#### WORKED EXAMPLE

**Worked example: JazzCash-style "Send Money" in Figma, without writing code.**

1. Lo-Fi on paper or in Figma: three frames — Home (boxes: Send, Bills, Load), Amount (text field + Next), Success (tick + Done).
2. Show the sketch to two classmates. If they cannot find Send, move that box to the top. This is the point of Lo-Fi: cheap changes.
3. In Figma, add a phone frame, rectangles, and labels. Duplicate the frame for Home, Amount and Success.
4. Prototype tab: Trigger = On Tap of the Send rectangle; Action = Navigate to Amount. On Tap of Next → Success.
5. Hand the laptop to a student. If they tap Bills by mistake, the navigation is still wrong — fix it before any programmer writes Java or Kotlin.

#### REMEMBER IT

**Lo-Fi = boxes. Hi-Fi = almost the real app. Prototype = clickable.** Figma can hold all three stages. Trigger = what the user does. Action = where the fake app goes.

#### COMMON MISTAKE

**Mistake:** Colouring a Lo-Fi wireframe, or thinking a Figma prototype is the finished coded app.

**Fix:** Lo-Fi stays black-and-white on purpose so people discuss structure, not taste in colours. A prototype tests ideas before code; developers still have to build the real system afterwards.

#### اردو میں سمجھیں

وائر فریم یو آئی کا بنیادی خاکہ ہے، رنگوں سے پہلے۔ لو فائی سیاہ و سفید ڈبے ہیں؛ ہائی فائی تقریباً حتمی شکل ہے، رنگ، فونٹ اور کلک ہونے والے بٹن کے ساتھ۔ فگما (Figma) مفت ویب ٹول ہے: فریم بنانا، شکل اور متن ڈالنا، فریم کاپی کرو۔ پروٹوٹائپ کوڈ سے پہلے کا چلنے والا نمونہ ہے جس میں ٹرگر (On Tap) اور ایکشن (اگلی اسکرین) جوڑے جاتے ہیں۔

#### CHECK YOURSELF

1. What does a Lo-Fi wireframe deliberately leave out?
2. Where does Figma run?
3. In Figma prototyping, what is a trigger? Give the source example.

Answers: 1. Colours and detailed graphics (it is structure only). 2. Directly in a web browser. 3. The user event that starts a move between screens, e.g. "On Tap".

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Next lecture: 1.11 Test and Evaluate HCI (Evaluation Methods)